

Intermediate Microeconomics

Week 11b: Imperfect Competition

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Oligopoly

- ❑ A market with relatively **few** firms but more than one
- ❑ Possibility of **strategic interaction** among firms
- ❑ **Difficult** to predict exactly the possible outcomes for price and output



Pricing Under Homogeneous Oligopoly

- We will assume that the market is *perfectly competitive on the demand side*
 - there are many buyers, each of whom is a price taker

- We will assume that the good obeys the law of *one price*
 - this assumption will be relaxed when product differentiation is discussed

Pricing Under Homogeneous Oligopoly

- We will assume that there is a *relatively small number of identical firms* (n)
 - we will initially start with n fixed, but later allow n to vary through entry and exit in response to firms' profitability

- The output of each firm is q_i ($i=1,\dots,n$)
 - symmetry in costs across firms will usually require that these outputs are equal

Pricing Under Homogeneous Oligopoly

- The inverse demand function for the good shows the price that buyers are willing to pay for any particular level of industry output

$$P = f(Q) = f(q_1 + q_2 + \dots + q_n)$$

- Each firm's goal is to maximize profits

$$\pi_i = f(Q)q_i - C_i(q_i)$$

$$\pi_i = f(q_1 + q_2 + \dots + q_n)q_i - C_i$$

Oligopoly Pricing Models

- The quasi-competitive model assumes price-taking behavior by all firms
 - P is treated as fixed
- The cartel model assumes that firms can collude perfectly in choosing industry output and P

Oligopoly Pricing Models

- The Cournot model assumes that firm i treats firm j 's output as fixed in its decisions
 - $\partial q_j / \partial q_i = 0$

- The conjectural variations model assumes that firm j 's output will respond to variations in firm i 's output
 - $\partial q_j / \partial q_i \neq 0$

Quasi-Competitive Model

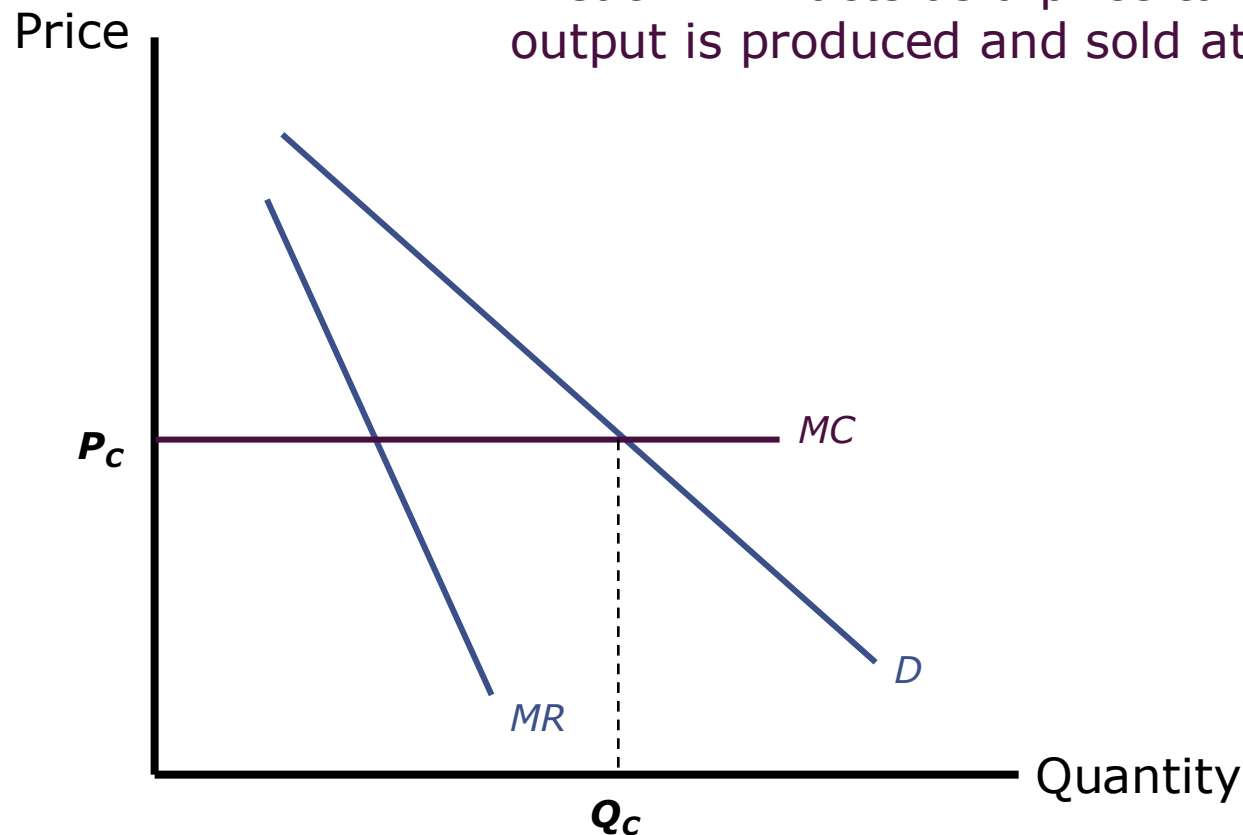
- Each firm is assumed to be a price taker
- The first-order condition for profit-maximization is

$$\begin{aligned}\partial\pi_i/\partial q_i &= P - (\partial C_i/\partial q_i) = 0 \\ P &= MC_i(q_i) \quad (i=1,\dots,n)\end{aligned}$$

- Along with market demand, these n supply equations will ensure that this market ends up at the short-run competitive solution

Quasi-Competitive Model

If each firm acts as a price taker, $P = MC_i$ so Q_C output is produced and sold at a price of P_C



Bertrand Model

- Two identical firms
 - Producing **identical** products at a constant $MC = c$
 - Choose prices p_1 and p_2 **simultaneously**
 - Single period of competition
 - How Sales get split
 - All sales go to the firm with the lowest price
 - Sales are **split evenly** if $p_1 = p_2$
-

Bertrand Model: The **Only** Pure-strategy Nash equilibrium

- The **Only** Pure-strategy Nash equilibrium:

$$p_1^* = p_2^* = c$$

- **Both** firms are playing a **best response** to each other
- **Neither** firm has an incentive to **deviate** to some other strategy

- A formal proof should verify that all other cases are not Nash equilibrium

- Let's focus on cases where $p_1 \leq p_2$
 - Three cases: $p_1^* < c$, $p_1^* > c$, $p_1^* = c$
-

Bertrand Model: The **Only** Pure-strategy Nash equilibrium

- If $p_1 < c$ (and $p_1 \leq p_2$)
 - Profit would be negative, should deviate to $p_1 = c$
 - If $p_1 > c$ (and $p_1 \leq p_2$)
 - Firm 2 could gain by **undercutting** the price of firm 1 and captures all the market
 - If $p_1 = c$ (and $p_1 \leq p_2$)
 - If $p_1 < p_2$, then firm 1 can raise price **slightly over** c but still lower than p_2 , and earn higher profit (because it still gets the whole market)
 - The **Only** Pure-strategy Nash equilibrium:
 $p_1^* = p_2^* = c$
-

Bertrand Model

- For any number of firms $n \geq 2$
 - The same result
 - Nash equilibrium of the n -firm Bertrand game is $p_1^* = p_2^* = \dots = p_n^* = c$
 - The Bertrand paradox
 - The Nash equilibrium of the Bertrand model is the same as the perfectly competitive outcome **even though there are only two firms**
 - Price is set to marginal cost
 - Firms earn zero profit
-

☐ The Bertrand paradox

- General : holds for ***any c*** and ***any downward-sloping*** demand curve
 - ***Not*** general: ***can be undone*** by changing assumptions:
 - ☐ Choosing quantity rather than price
 - ☐ Facing capacity constraint
 - ☐ Products slightly differentiated (not perfect substitute)
 - ☐ Repeated interaction
-

Cartel Model

- The assumption of price-taking behavior may be inappropriate in oligopolistic industries
 - each firm can recognize that its output decision will affect price

- An alternative assumption would be that firms act as a group and coordinate their decisions so as to achieve monopoly profits

Cartel Model

- In this case, the cartel acts as a multiplant monopoly and chooses q_i for each firm so as to maximize total industry profits

$$\pi = PQ - [C_1(q_1) + C_2(q_2) + \dots + C_n(q_n)]$$

- If write everything in terms of q_i

$$\pi = f(q_1 + q_2 + \dots + q_n)[q_1 + q_2 + \dots + q_n] - \sum_{i=1}^n C_i(q_i)$$

Cartel Model

- The first-order conditions for a maximum are that

$$\frac{\partial}{\partial q_i} \left(\sum_{j=1}^n \pi_j \right) = P(Q) + P'(Q) \sum_{j=1}^n q_j - C'(q_i) = 0 \quad \text{for } i = 1, \dots, n$$

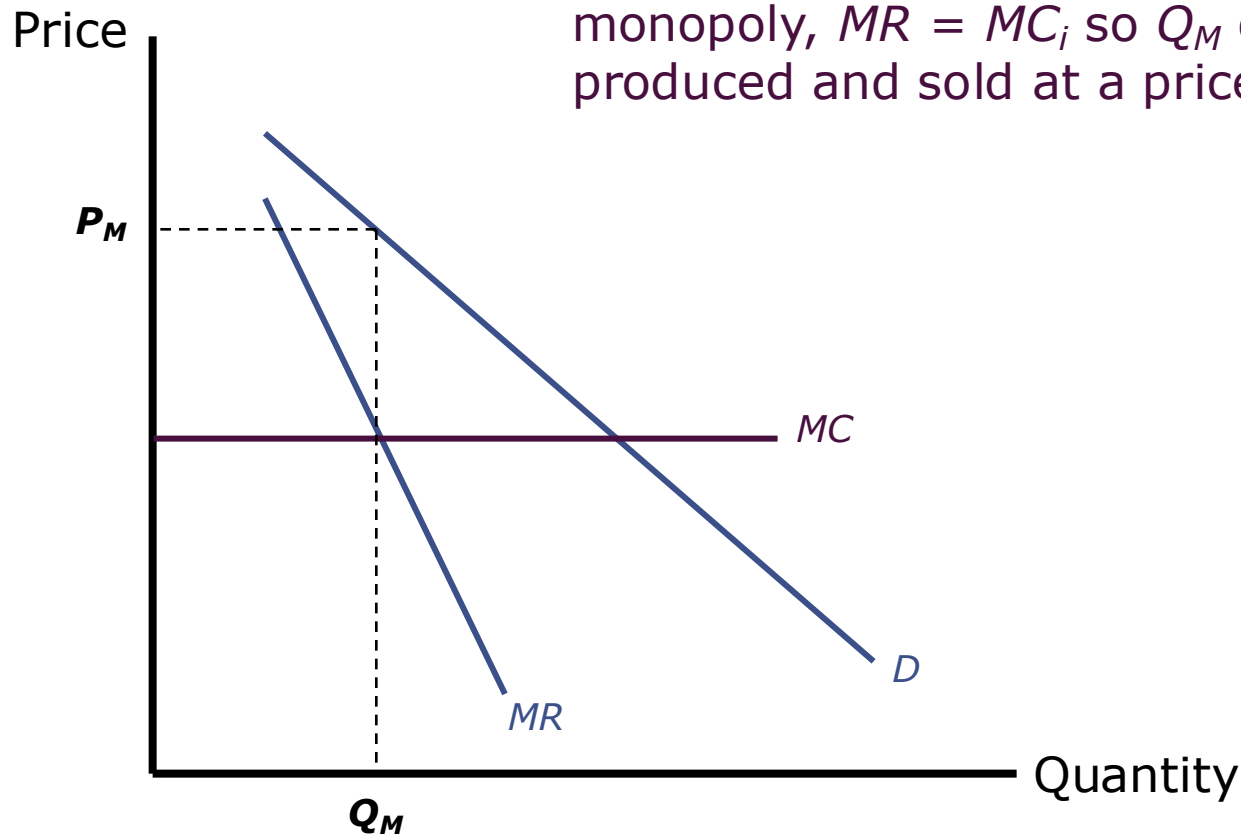
- This implies that

$$MR(Q) = MC_i(q_i)$$

- At the profit-maximizing point, marginal revenue will be equal to each firm's marginal cost

Cartel Model

If the firms form a group and act as a monopoly, $MR = MC_i$ so Q_M output is produced and sold at a price of P_M



Cartel Model

- There are three problems with the cartel solution
 - these monopolistic decisions may be illegal
 - it requires that the directors of the cartel know the market demand function and each firm's marginal cost function
 - the solution may be unstable
 - each firm has an incentive to expand output because $P > MC_i$

Cournot Model

- Each firm recognizes that its own decisions about q_i affect price
 - $\partial P / \partial q_i \neq 0$

- However, each firm believes that its decisions do not affect those of any other firm
 - $\partial q_j / \partial q_i = 0$ for all $j \neq i$

Cournot Model

- Firm i 's profit = total revenue – total cost

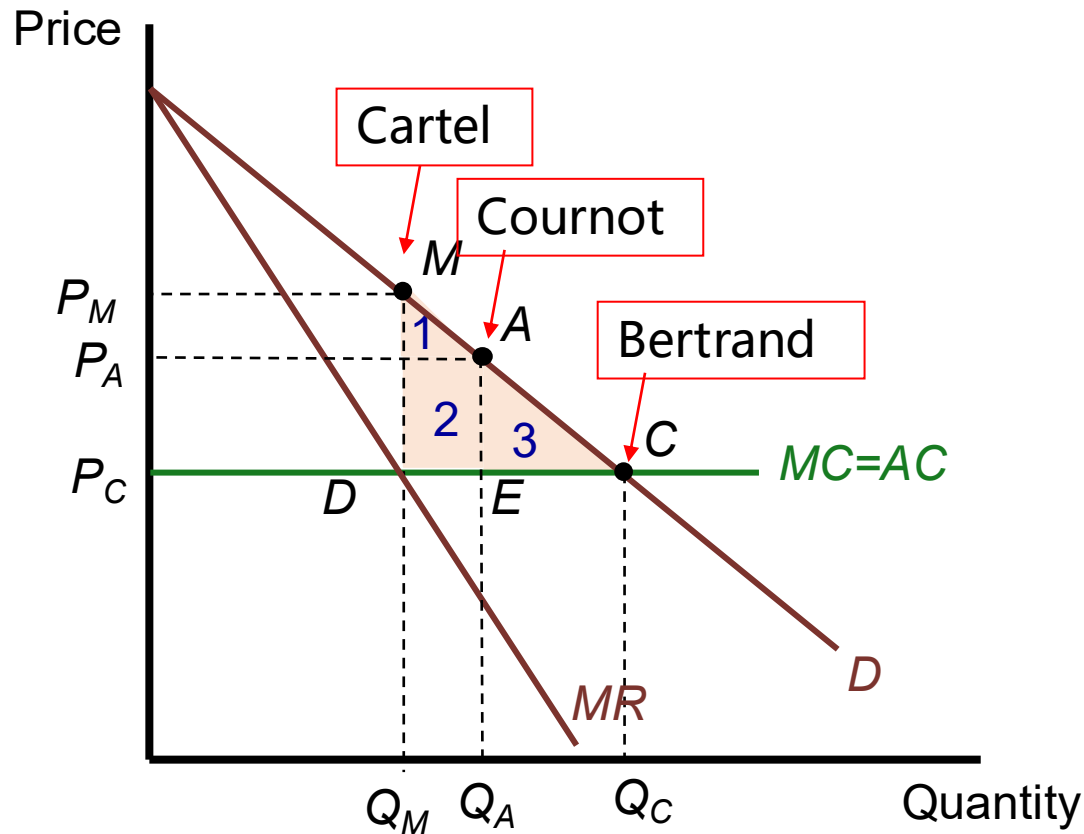
$$\pi_i = P(Q)q_i - C_i(q_i)$$

- First-order conditions for profit maximization:

$$\frac{\partial \pi_i}{\partial q_i} = \underbrace{P(Q) + P'(Q)q_i}_{\text{MR}} - \underbrace{C'_i(q_i)}_{\text{MC}} = 0$$

- Maximize profit where $MR_i = MC_i$
 - the firm assumes that changes in q_i affect its total revenue only through their direct effect on market price

Bertrand vs. Cournot vs. Cartel



- In Cournot game, industry profits
 - Lower than in the cartel model ($P_A A E P_C < P_M M D P_C$)
- DWL
 - Smaller in the Cournot model (3) than in the cartel situation (1+2+3)

Varying the Number of Cournot Firms

□ The Cournot model

- Can represent the whole range of outcomes by varying the number of firms
- $n = \infty \Rightarrow$ perfect competition
- $n = 1 \Rightarrow$ perfect cartel / monopoly

□ n identical firms

- Same cost function $C(q_i)$
- In equilibrium, each produces $q_i = Q/n$

Varying the Number of Cournot Firms

- Difference between price and marginal cost:

$$P'(Q)Q/n$$

- The wedge term disappears as n grows large; firms become infinitesimally small – price takers
 - Price approaches marginal cost
 - Market outcome approaches the perfectly competitive one
- As n decreases to 1: the Cournot outcome approaches that of a perfect cartel

Conjectural Variations Model

- In markets with only a few firms, we can expect there to be strategic interaction among firms
- One way to build strategic concerns into our model is to consider the assumptions that might be made by one firm about the other firm's behavior

Conjectural Variations Model

- For each firm i , we are concerned with the assumed value of $\partial q_j / \partial q_i$ for $i \neq j$
- because the value will be speculative, models based on various assumptions about its value are termed conjectural variations models
 - they are concerned with firm i 's conjectures about firm j 's output variations

Conjectural Variations Model

- The first-order condition for profit maximization becomes

$$\frac{\partial \pi_i}{\partial q_i} = P + q_i \left[\frac{\partial P}{\partial q_i} + \sum_{j \neq i} \frac{\partial P}{\partial q_j} \cdot \frac{\partial q_j}{\partial q_i} \right] - MC_i(q_i) = 0$$

The firm must consider how its output decisions will affect price in two ways

- directly
- indirectly through its effect on the output decisions of other firms

Practice example: Natural Springs Duopoly

- Assume that there are two owners of natural springs
 - each firm has no production costs
 - each firm has to decide how much water to supply to the market

- The demand for spring water is given by the linear demand function

$$Q = q_1 + q_2 = 120 - P$$

Natural Springs Duopoly

- In a Bertrand model, what are the market price and the quantity supplied?

Natural Springs Duopoly

- In a Cartel model, what are the market price and the quantity supplied?

Cournot's Natural Springs Duopoly

- In a Cournot model, what are the market price and the quantity supplied?

Cournot's Natural Springs Duopoly

- These first-order equations are called best-response functions (or reaction functions)
 - show how each firm reacts to the other's output level
- We can solve the reaction functions simultaneously to find that

$$q_1 = q_2 = 40$$

$$P = 120 - (q_1 + q_2) = 40$$

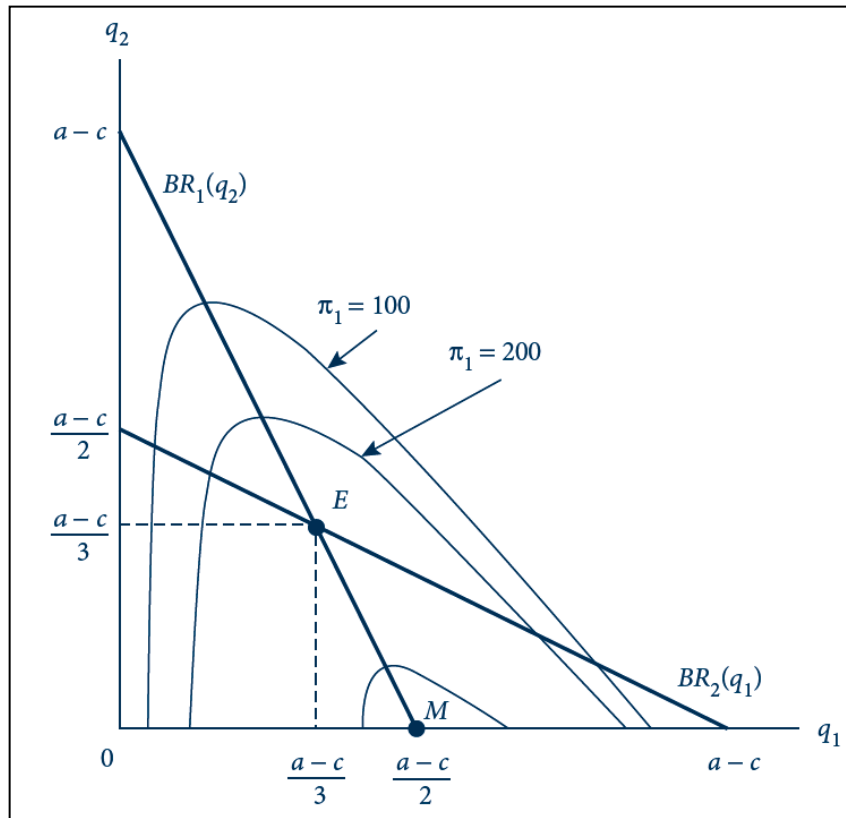
$$\pi_1 = \pi_2 = Pq_1 = Pq_2 = 1,600$$

- Note that the Cournot equilibrium falls between the quasi-competitive model and the cartel model

EXAMPLE 15.2 Cournot Best-Response Diagrams

- Solve for the Nash equilibrium using graphical methods
 - Graph the intercepts of the best-response functions
 - Intersection between the best responses is the Nash equilibrium
- An isoprofit curve for firm 1
 - Is the locus of quantity pairs providing it with the same profit level

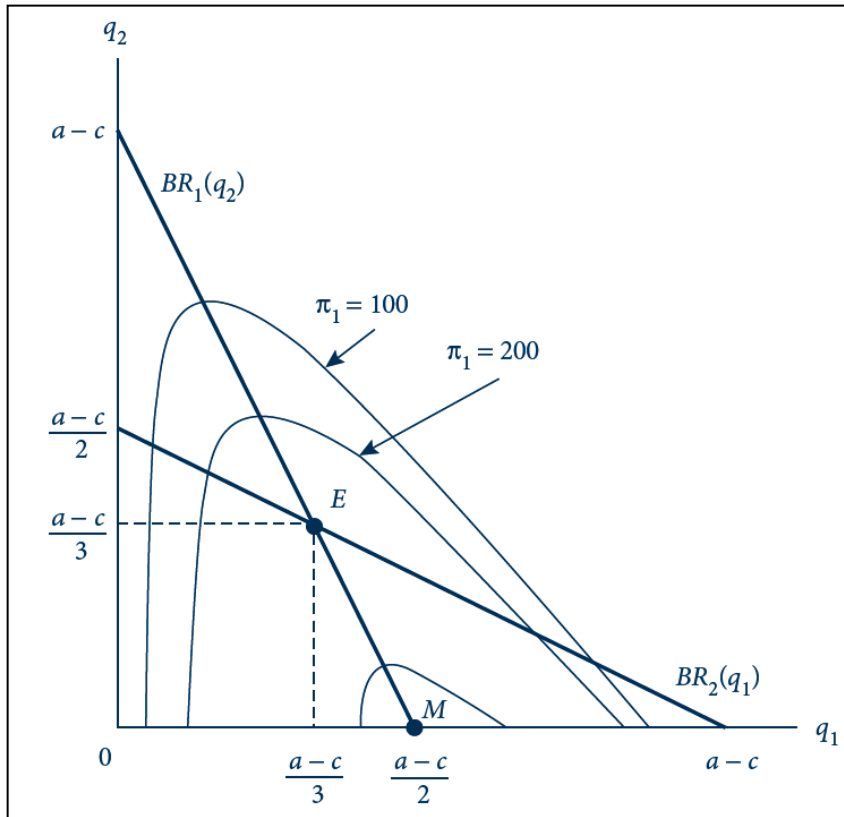
Best-Response Diagram for Cournot Duopoly



Demand: $P(Q) = a - Q$
Cost: $C_i(q_i) = cq_i$

- Solve for the Cournot firms' best response functions.

Best-Response Diagram for Cournot Duopoly



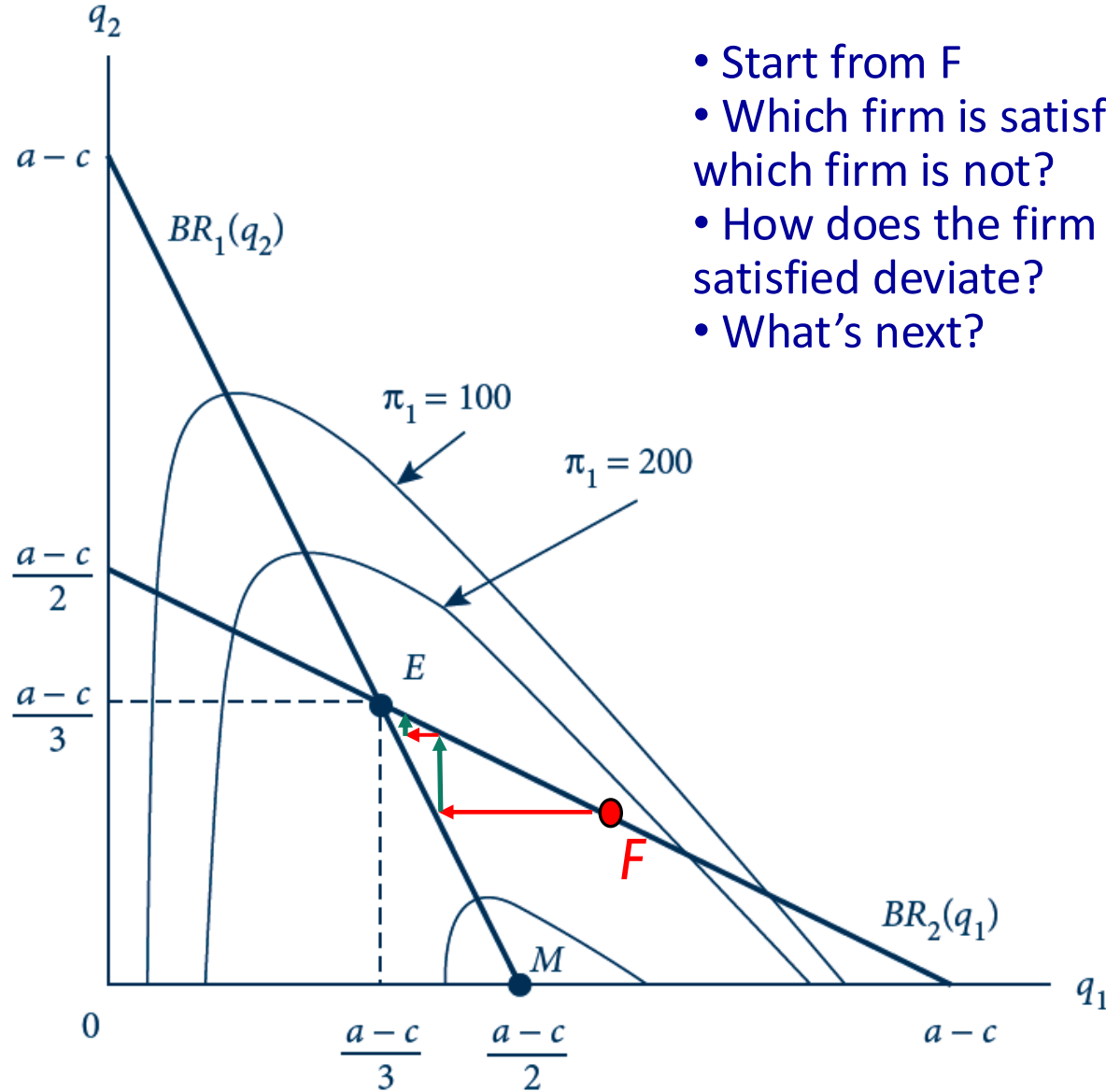
Demand: $P(Q) = a - Q$
 Cost: $C_i(q_i) = cq_i$

- Firms' best responses are drawn as thick lines;
 - Their intersection (E) is the Nash equilibrium of the Cournot game.

$$q_1 = \frac{a - q_2 - c}{2}$$

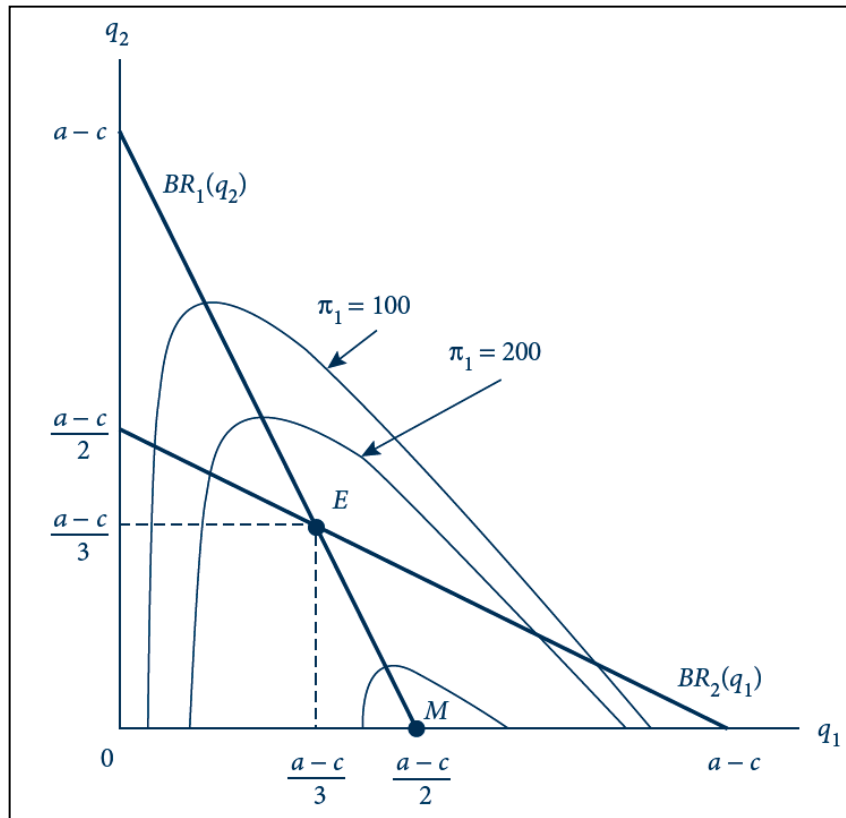
$$q_2 = \frac{a - q_1 - c}{2}$$

Best-Response Diagram for Cournot Duopoly



- Start from F
- Which firm is satisfied at F , which firm is not?
- How does the firm that's not satisfied deviate?
- What's next?

Best-Response Diagram for Cournot Duopoly



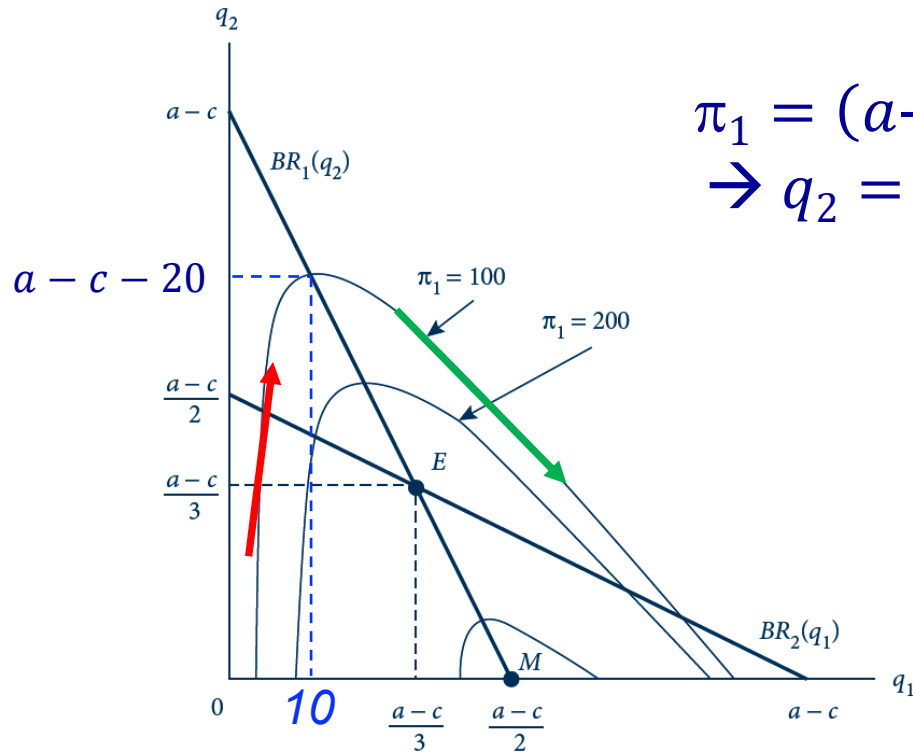
Demand: $P(Q) = a - Q$
 Cost: $C_i(q_i) = cq_i$

- Firms' best responses are drawn as thick lines;
 - Their intersection (E) is the Nash equilibrium of the Cournot game.

$$q_1 = \frac{a - q_2 - c}{2} \qquad q_2 = \frac{a - q_1 - c}{2}$$

- An iso-profit curve for firm 1
 - Is the **locus** of quantity pairs providing it with the same profit level

Iso-profit curve: inverse U-shape



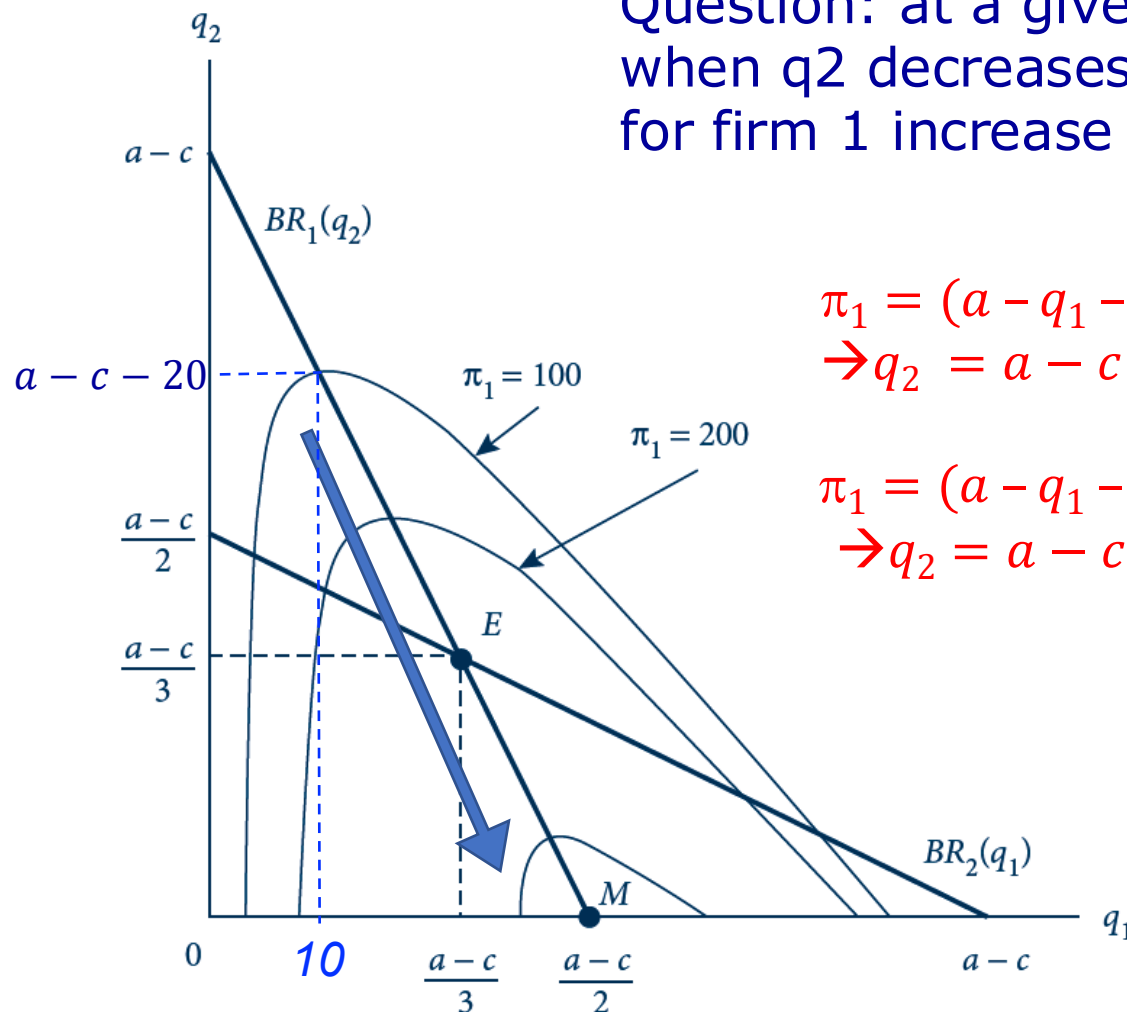
$$\pi_1 = (a - q_1 - q_2)q_1 - cq_1 = 100$$

$$\rightarrow q_2 = a - c - (q_1 + 100/q_1)$$

- As q_1 was close to 0 and q_1 increases, $100/q_1$ dominates, and $q_1 + 100/q_1$ decreases if $q_1 < 10$
 - So if $q_1 < 10$, q_2 must be increasing to keep profit constant at 100
- As q_1 increases further (> 10), q_1 will begin to dominate, and $q_1 + 100/q_1$ increases
 - So q_2 must be decreasing to keep profit constant at 100

Iso-profit curve

Question: at a given level of q_1 , when q_2 decreases, does profit for firm 1 increase or decrease?



$$\pi_1 = (a - q_1 - q_2 - c)q_1 = 100$$

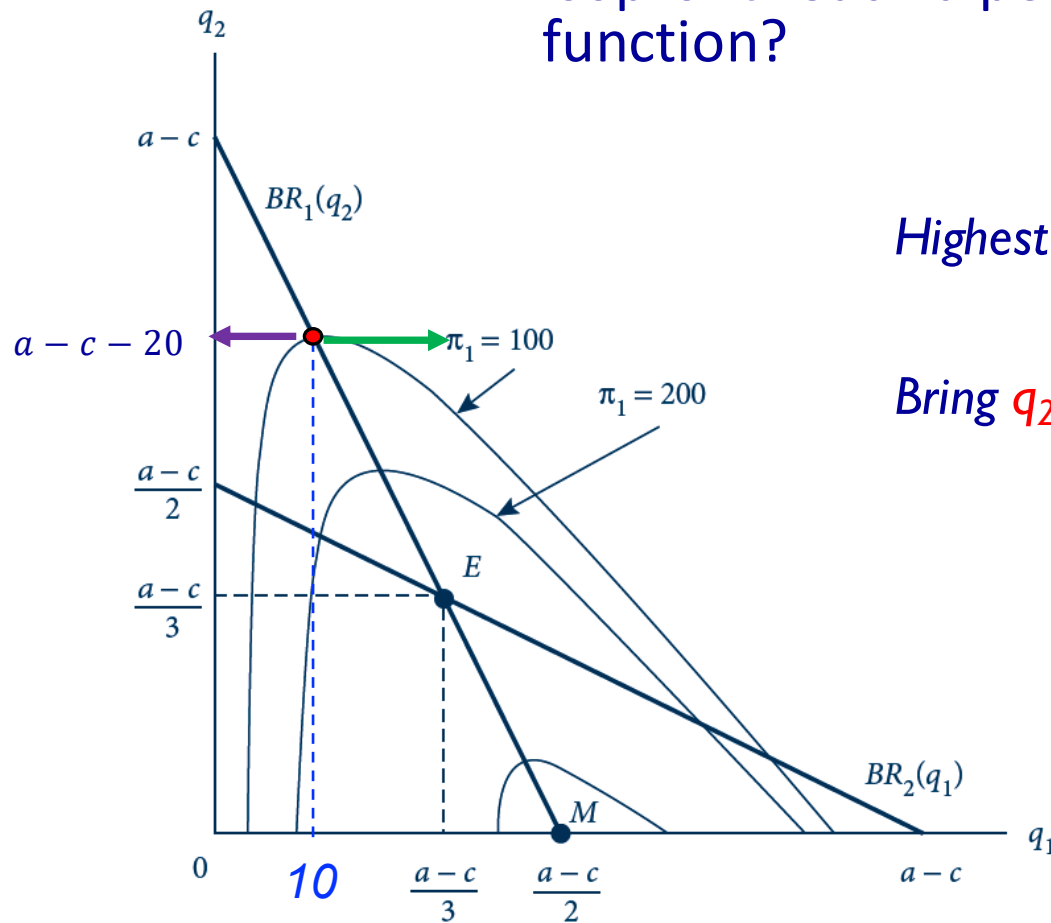
$$\rightarrow q_2 = a - c - q_1 - 100/q_1$$

$$\pi_1 = (a - q_1 - q_2 - c)q_1 = 200$$

$$\rightarrow q_2 = a - c - q_1 - 200/q_1$$

- As profit increases from 100 to 200 to yet higher levels, the associated isoprofits shrink down to the monopoly point, which is the highest isoprofit on the diagram.

Question: Why does firm 1's individual isoprofit reach a peak on its best-response function?



Highest q_2 On this curve:

$$q_1^* = 10, q_2^* = a - c - 20$$

Bring q_2^* to the best-response curve:

$$\begin{aligned} q_1 &= \frac{a - q_2 - c}{2} \\ &= \frac{a - c - (a - c - 20)}{2} \\ &= 10 \end{aligned}$$

Intuition: On firm 1's best-response function, for a given level of q_2

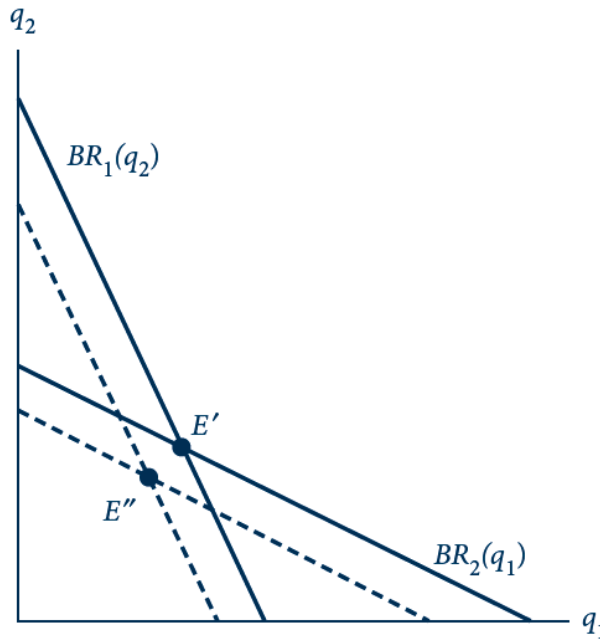
- If firm 1 increases its output q_1 , profit will decrease.
- If firm 1 decreases its output q_1 , profit will also decrease.

Hence, the point on the best-response function is at the peak of the isoprofit curve.

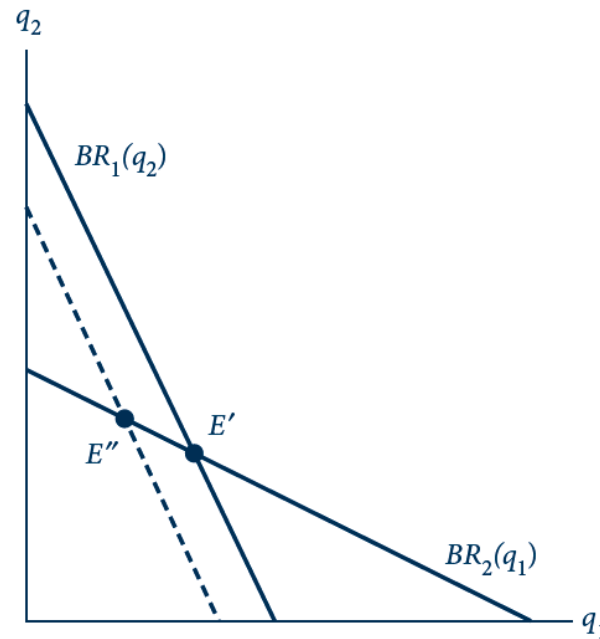
Best-response diagrams

$$q_1 = \frac{a - q_2 - c}{2}$$

$$q_2 = \frac{a - q_1 - c}{2}$$



(a) Increase in both firms' marginal costs



(b) Increase in firm 1's marginal cost

- Panel (a) depicts an increase in both firms' marginal costs, c , shifting their best responses *inward*.
- If marginal costs are different as in Panel (b), output q_1 is lower, q_2 is higher.
- What about an increase in the preference parameter, a ?

Practice example:

- Let c_i be the constant marginal and average cost for firm i (so that firms may have different marginal costs). Suppose demand is given by $P=1-Q$.
- 1. Calculate the Nash equilibrium quantities assuming there are two firms in a Cournot market. Also compute market output, market price, firm profits, industry profits, consumer surplus, and total welfare.
- 2. Represent the Nash equilibrium on a best-response function diagram. Show how a reduction in firm 1's cost would change the equilibrium. Draw a representative isoprofit for firm 1.

Prices or Quantities?

- Bertrand model - price competition
 - Discontinuous jump from monopoly to perfect competition if just two firms enter
 - Additional entry beyond two has no additional effect on the market outcome
- Cournot model - quantity competition
 - Industry grows more competitive as the number n of firms entering the market increases

Capacity Constraints

- For the Bertrand model to generate the Bertrand paradox
 - Firms must have unlimited capacity
 - More realistically, firms may not have an unlimited ability to meet all demand
- Starting from equal prices, if a firm lowers its price the slightest amount, then its demand essentially doubles. The firm can satisfy this increased demand because it has no capacity constraints.
- If the undercutting firm could not serve all the demand because of capacity constraints, that would leave some residual demand for the higher-priced firm and would decrease the incentive to undercut.

Capacity Constraints

- Two-stage game
 - Firms build capacity in the first stage
 - Firms choose prices p_1 and p_2 in the second stage
 - Sales of firms cannot exceed the capacity chosen in the first stage
 - If the cost of building capacity is sufficiently high
 - Equilibrium - the same as the Nash equilibrium of the Cournot model
 - As if firms choose quantities rather than price.

Product Differentiation

- To avoid the Bertrand paradox
 - Assume that firms produce differentiated products
- Market
 - A group of closely related products
 - That are more substitutable among each other
 - As measured by cross-price elasticities
 - Than with goods outside the group

Bertrand competition with differentiated products

- There are n firms competing in a particular market
 - Each product has its own attributes, a_i
- The product's attributes affect its demand

$$q_i(p_i, P_{-i}, a_i, A_{-i})$$

- Where P_{-i} is a list of all other firms' prices
- And A_{-i} is a list of the attributes of other firms' products

Bertrand competition with differentiated products

□ Firm i 's

■ Total cost: $C_i(q_i, a_i)$

■ Profit: $\pi_i = p_i q_i - C_i(q_i, a_i)$

□ First-order conditions for a maximum:

$$\frac{\partial \pi_i}{\partial p_i} = q_i + p_i \frac{\partial q_i}{\partial p_i} - \frac{\partial C_i}{\partial q_i} \cdot \frac{\partial q_i}{\partial p_i} = 0$$

$$\frac{\partial \pi_i}{\partial a_i} = p_i \frac{\partial q_i}{\partial a_i} - \frac{\partial C_i}{\partial a_i} - \frac{\partial C_i}{\partial q_i} \cdot \frac{\partial q_i}{\partial a_i} = 0$$

Bertrand competition with differentiated products

□ First-order conditions for a maximum:

$$\frac{\partial \pi_i}{\partial p_i} = \boxed{q_i + p_i \frac{\partial q_i}{\partial p_i}} - \boxed{\frac{\partial C_i}{\partial q_i} \cdot \frac{\partial q_i}{\partial p_i}} = 0$$

Blue box: marginal revenue from an increase in price.

Red box: cost savings associated with the reduced sales that accompany an increased price.

Regarding the choice of a_i is more complex, let's take a look at the next two examples.

EXAMPLE 15.4 Toothpaste as a Differentiated Product

- Two firms produce toothpaste
 - One a green gel and the other a white paste
 - Suppose that production is costless
- Demand for product i , $q_i = a_i - p_i + p_j/2$
 - The goods are gross substitutes (positive coefficient on p_j)
 - Suppose that attribute a_i is an endowment rather than a choice variable for the firm.

EXAMPLE 15.4 Toothpaste as a Differentiated Product

- Two firms produce toothpaste
 - One a green gel and the other a white paste
 - Suppose that production is costless
- Demand for product i , $q_i = a_i - p_i + p_j/2$
- Firm i 's profit: $\pi_i = p_i q_i - C_i(q_i) = p_i(a_i - p_i + p_j/2)$
 - Where $C_i(q_i) = 0$ for simplicity
 - First-order condition for profit maximization

$$\frac{\partial \pi_i}{\partial p_i} = a_i - 2p_i + p_j/2 = 0$$

EXAMPLE 15.4 Toothpaste as a Differentiated Product

- Best-response functions

$$p_1 = \frac{1}{2} \left(a_1 + \frac{p_2}{2} \right), \quad p_2 = \frac{1}{2} \left(a_2 + \frac{p_1}{2} \right)$$

- Nash equilibrium prices

$$p_i^* = \frac{8}{15} a_i + \frac{2}{15} a_j$$

- Firm i 's equilibrium price is not only increasing in its own attribute, a_i , but also in the other product's attribute, a_j .
- An increase in a_j causes firm j to increase its price, which increases firm i 's demand and thus the price firm i charges.

EXAMPLE 15.4 Toothpaste as a Differentiated Product

- Best-response functions

$$p_1 = \frac{1}{2} \left(a_1 + \frac{p_2}{2} \right), \quad p_2 = \frac{1}{2} \left(a_2 + \frac{p_1}{2} \right)$$

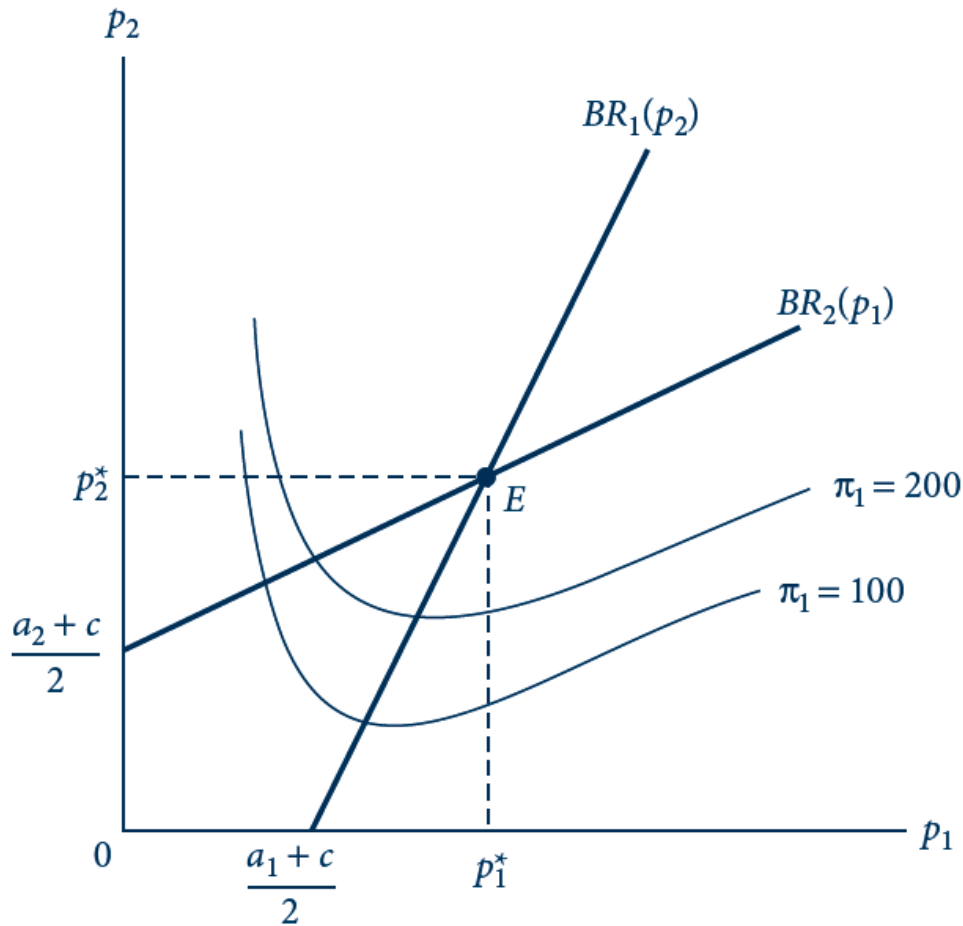
- Nash equilibrium prices

$$p_i^* = \frac{8}{15} a_i + \frac{2}{15} a_j$$

- Profits

$$\pi_i^* = \left(\frac{8}{15} a_i + \frac{2}{15} a_j \right)^2$$

FIGURE 15.4 Best Responses for Bertrand Model with Differentiated Products

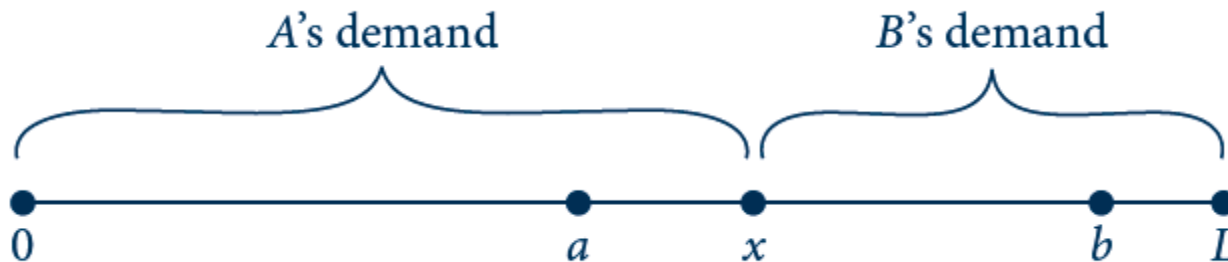


Firm' best responses are drawn as thick lines; their intersection (E) is the Nash equilibrium.

EXAMPLE 15.5 Hotelling's Beach

- Ice cream stands located on a beach
 - We will take the locations of the ice cream stands as given.
 - Demanders are located uniformly along the beach
 - One at each unit of beach
 - Ice cream cones are costless to produce
 - But carrying them back to one's place on the beach results in a cost of td^2
 - t = temperature
 - d = distance

FIGURE 15.5 Hotelling's Beach



Ice cream stands A and B are located at points a and b along a beach of length L . The consumer who is indifferent between buying from the two stands is located at x . Consumers to the left of x buy from A and to the right buy from B.

EXAMPLE 15.5 Hotelling's Beach

- A person located at point x will be indifferent between stands A and B if

$$p_A + t(x - a)^2 = p_B + t(b - x)^2$$

- Where p_A and p_B are the prices charged by each stand, and $t(x-a)^2$ is the transportation cost.
- Solving for x we get
$$x = \frac{b + a}{2} + \frac{p_B - p_A}{2t(b - a)}$$
 - If the two stands charge an equal price, the indifferent consumer is located midway between a and b

EXAMPLE 15.5 Hotelling's Beach

- Consumers 0 to x buy from A;

$$q_A(p_B, p_A, a, b) = x = \frac{b+a}{2} + \frac{p_B - p_A}{2t(b-a)}$$

- the remaining $L-x$ consumers buy from B.

$$q_B(p_B, p_A, b, a) = L - x = L - \frac{b+a}{2} + \frac{p_A - p_B}{2t(b-a)}$$

- Solve for Nash Equilibrium price and profits for the two firms.

EXAMPLE 15.5 Hotelling's Beach

- The Nash equilibrium prices:

$$p_A^* = \frac{t}{3}(b-a)(2L+a+b)$$

$$p_B^* = \frac{t}{3}(b-a)(4L-a-b)$$

- Profits for the two firms:

$$\pi_A^* = \frac{t}{18}(b-a)(2L+a+b)^2$$

$$\pi_B^* = \frac{t}{18}(b-a)(4L-a+b)^2$$